

Required Number of Points in L_2 Marcinkiewicz-Zygmund inequalities

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L_2 Marcinkiewicz-Zygmund (MZ) inequalities

Definition

[Marcinkiewicz, Zygmund 1937]

Let $\bullet$ Ω be a domain,

- $\bullet$ μ a measure on Ω ,
- $\bullet$ $L_2 = L_2(\Omega, \mu)$ the space of square-integrable functions, and
- $\bullet$ $\mathcal{F}$ an m -dimensional function space.

Points $x_1, \dots, x_n \in \Omega$ and weights $w_1, \dots, w_n > 0$ fulfill an L_2 MZ inequality with constants $0 < (1 - \varepsilon) \leq (1 + \varepsilon) < \infty$, iff

$$(1 - \varepsilon) \|f\|_{L_2}^2 \leq \sum_{i=1}^n w_i |f(x_i)|^2 \leq (1 + \varepsilon) \|f\|_{L_2}^2 \quad \forall f \in \mathcal{F}. \quad (\text{MZ})$$

Example of L_2 MZ inequalities on $\mathbb{T}$

Example (equidistant points on $\mathbb{T}$)

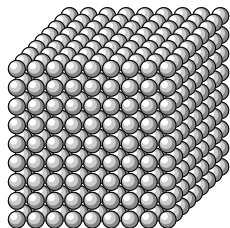
Let • $L_2 = L_2(\mathbb{T}, dx)$, $n \in \mathbb{N}$, and

• $\mathcal{F} = \text{span}\{\exp(2\pi i k \cdot)\}_{k=0, \dots, n-1}$.

Then

$$\|f\|_{L_2}^2 = \frac{1}{n} \sum_{i=1}^n |f(\frac{i}{n})|^2 \quad \forall f \in \mathcal{F}.$$

Note: $\varepsilon = 0$ and $n = m$



point grid in $d = 3$

Known upper bounds

--- Christoffel sampling

[Cohen, Davenport, Leviatan 2017]

$$n = \left\lceil \frac{6m \log m}{\varepsilon^2} \right\rceil$$

--- BSS subsampling

[B., Schäfer, T. Ullrich 2023]

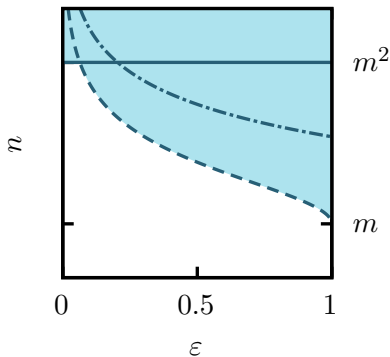
$$n = \left\lceil m \frac{1 + \sqrt{1 - \varepsilon^2}}{1 - \sqrt{1 - \varepsilon^2}} \right\rceil \leq \frac{5m}{\varepsilon^2}$$

— Carathéodory subsampling

[B., Kämmerer, Pozharska, Schäfer, T. Ullrich 2025],

[Schäfer, T. Ullrich 2025]

$$n = m^2$$



upper bounds for
 $\dim(\mathcal{F}) = m = 998$

Nonuniform weighting is necessary

Example

[Chkifa, Dolbeault, Krieg, M. Ullrich 2026]

Let $\bullet \Omega = \{0, 1\}$, $\mu(\{0\}) = \delta \in (0, 1)$, and $\mu(\{1\}) = 1 - \delta$,

$\bullet \mathcal{F} = \{f: \Omega \rightarrow \mathbb{R}\}$ and $0 \leq \varepsilon < 1$.

Assuming points $x_1, \dots, x_n$ and a uniform scaling $w > 0$ exist with

$$(1 - \varepsilon) \|f\|_{L_2}^2 \leq w \sum_{i=1}^n |f(x_i)|^2 \leq (1 + \varepsilon) \|f\|_{L_2}^2 \quad \forall f \in \mathcal{F}.$$

We have

- $\bullet \varepsilon < 1$ implies at least one $x_i = 0$.
- $\bullet f = \mathbb{1}_{\{0\}}$ and upper MZ inequality implies $w \leq w |\{i : x_i = 0\}| \leq (1 + \varepsilon)\delta$.
- $\bullet f \equiv 1$ and lower MZ inequality implies $(1 - \varepsilon) \leq wn$

Thus,

$$n \geq \frac{1 - \varepsilon}{1 + \varepsilon} \frac{1}{\delta} \rightarrow \infty \quad \text{for } \delta \rightarrow 0.$$

Trivial lower bound

Dimension argument

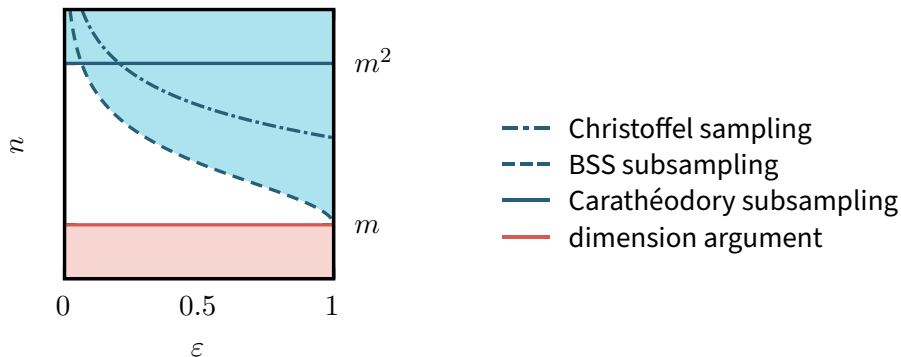
Let the number of points be $n < \dim(\mathcal{F}) = m$. Then there exists $f \in \mathcal{F}$ s.t.

- $\|f\|_{L_2} \neq 0$, but
- $f(x_1) = \dots = f(x_n) = 0$.

As L_2 -MZ requires $1 - \varepsilon > 0$, this gives the lower bound

$$n \geq \dim(\mathcal{F}) = m.$$

Dimension argument



number of points for L_2 MZ inequalities with constants $1 - \varepsilon$ and $1 + \varepsilon$
for $\dim(\mathcal{F}) = m = 998$

Goal: Construct hard to discretize function space $\mathcal{F}$ on Ω with measure μ .

In other words, for $n \in \mathbb{N}$ and arbitrary $x_1, \dots, x_n \in \Omega, w_1, \dots, w_n > 0$ satisfying

$$(1 - \varepsilon) \|f\|_{L_2}^2 \leq \sum_{i=1}^n w_i |f(x_i)|^2 \leq (1 + \varepsilon) \|f\|_{L_2}^2 \quad \forall f \in \mathcal{F} \quad (\text{MZ})$$

we prove a lower bound on ε in terms of n .

Frame function space

Definition

The vectors $\Phi = \{\varphi_1, \dots, \varphi_N\} \subset \mathbb{C}^m$ form a unit-norm tight frame (UNTF), iff

$$\|\varphi_1\|_2 = \dots = \|\varphi_N\|_2 = 1$$

and

$$A\|\mathbf{a}\|_2^2 \leq \sum_{i=1}^N |\langle \mathbf{a}, \varphi_i \rangle|^2 \leq B\|\mathbf{a}\|_2^2 \quad \forall \mathbf{a} \in \mathbb{C}^m$$

with $A = B \neq 0$.

Frame function space

Proposition (frame function space construction)

- Let
- $\Phi = \{\varphi_1, \dots, \varphi_N\} \subset \mathbb{C}^m$ be a UNTF,
 - $\Omega = \{1, \dots, N\}$ be the domain, μ the counting measure, and

$$\mathcal{F}_\Phi := \left\{ f_{\mathbf{a}}(i) = \sqrt{\frac{m}{N}} \langle \mathbf{a}, \varphi_i \rangle : \mathbf{a} \in \mathbb{C}^m \right\}.$$

Then points $J \subseteq \{1, \dots, N\}$ and weights $w_i > 0$ satisfy (MZ) on $\mathcal{F}_\Phi$ iff

$$\frac{N}{m}(1 - \varepsilon) \|\mathbf{a}\|_2^2 \leq \sum_{i \in J} w_i |\langle \mathbf{a}, \varphi_i \rangle|^2 \leq \frac{N}{m}(1 + \varepsilon) \|\mathbf{a}\|_2^2 \quad \forall \mathbf{a} \in \mathbb{C}^m.$$

General theorem for UNTFs

Theorem

[B 2026]

- Let $\bullet \Phi = \{\varphi_1, \dots, \varphi_N\} \subset \mathbb{C}^m$ be a UNTF,
- $\bullet Q_i: \mathbb{C}^m \rightarrow \mathbb{C}^m, \mathbf{a} \mapsto \langle \mathbf{a}, \varphi_i \rangle \varphi_i - \frac{1}{m} \mathbf{a}$ be centered projectors,
 - $\bullet \mathbf{G}_\Phi = [\text{trace}(Q_i Q_j)]_{i,j=1}^N$ be the Gram matrix, and
 - $\bullet \gamma_\Phi := \lambda_{\min}(\mathbf{G}_\Phi|_{\mathbf{1}_N^\perp})$.

Suppose $J \subseteq \{1, \dots, N\}$ and $w_i > 0$ satisfy (MZ). Then

$$|J| \geq \frac{m\gamma_\Phi N(1 - \varepsilon^2)}{m\gamma_\Phi(1 - \varepsilon^2) + N\varepsilon^2}.$$

ETFs as maximizers

Theorem

[B 2026]

Let $\Phi = \{\varphi_1, \dots, \varphi_N\} \subset \mathbb{C}^m$ be a UNTF with $\gamma_\Phi > 0$.

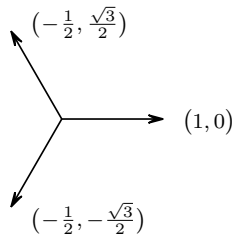
Then

$$N \leq m^2$$

and

$$\gamma_\Phi \leq \frac{N(m-1)}{m(N-1)}.$$

Equality holds iff Φ is an equiangular tight frame (ETF).



Mercedes Benz frame

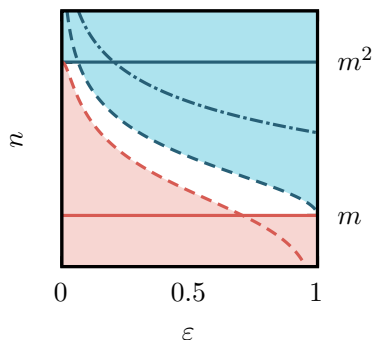
Maximal ETFs and Singer ETFs

Corollary (maximal ETFs)

[B 2026]

Suppose that a maximal ETF ($N = m^2$) exists in $\mathbb{C}^m$. Any set of n points and weights satisfying (MZ) on $\mathcal{F}_\Phi$ obeys

$$n \geq \frac{m^2(1 - \varepsilon^2)}{1 + m\varepsilon^2}.$$



- Christoffel sampling
- .- BSS subsampling
$$n = \left\lceil m \frac{1 + \sqrt{1 - \varepsilon^2}}{1 - \sqrt{1 - \varepsilon^2}} \right\rceil \leq \frac{5m}{\varepsilon^2}$$
- Carathéodory subsampling
$$n = m^2$$
- dimension argument
- - - maximal ETF construction

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Zauner's conjecture

[Zauner 1999]

Maximal ETFs exist in every dimension.

Maximal ETFs and Singer ETFs

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Zauner's conjecture

[Zauner 1999]

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Corollary (Singer ETFs)

[B 2026]

Let $m - 1$ be a prime power and Φ be the Singer ETF ($N = m^2 - m + 1$). Any set of n points and weights satisfying (MZ) on $\mathcal{F}_\Phi$ obeys

$$n \geq \frac{(m^2 - m + 1)(1 - \varepsilon^2)}{1 + (m - 1)\varepsilon^2}.$$

Graph-based construction

Proposition (Graph function space construction)

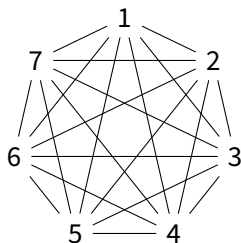
Let $\bullet \mathcal{K}$ be the complete graph

- $\bullet \mathcal{G} = (V, E, w)$ be a weighted subgraph
- $\bullet \mathbf{L}_{\mathcal{K}}$ and $\mathbf{L}_{\mathcal{G}}$ be their graph Laplacians, and

$$\mathcal{F}_{\mathcal{K}} := \left\{ f_{\mathbf{a}} : \binom{V}{2} \rightarrow \mathbb{R} : f_{\mathbf{a}}(\{u, v\}) = \frac{a_u - a_v}{\sqrt{|V|}}, \right. \\ \left. \mathbf{a} \in \mathbb{R}^{|V|} \text{ with } \sum_{u \in V} a_u = 0 \right\}.$$

Then points $J \subseteq \binom{V}{2}$ and weights $w_e > 0$ satisfy (MZ) iff

$$(1 - \varepsilon) \mathbf{L}_{\mathcal{K}} \preceq \mathbf{L}_{\mathcal{G}} \preceq (1 + \varepsilon) \mathbf{L}_{\mathcal{K}}.$$



complete graph on 7 nodes

Graph-based lower bound

Theorem

[B 2026]

The vectors $\{\frac{e_u - e_v}{\sqrt{2}} : \{u, v\} \in \binom{V}{2}\}$ form a UNTF for $\mathbf{1}_{|V|}^\perp \subset \mathbb{R}^{|V|}$.

The centered-projector Gram matrix G_K has eigenvalue zero on $\mathbf{1}_N$, and, on $\mathbf{1}_N^\perp$,

$$\frac{|V|}{4} \text{ with multiplicity } |V| - 1 \quad \text{and} \quad \frac{1}{2} \text{ with multiplicity } \frac{|V|(|V| - 3)}{2}.$$

Consequently, $\gamma_K = \frac{1}{2}$ for $|V| \geq 4$.

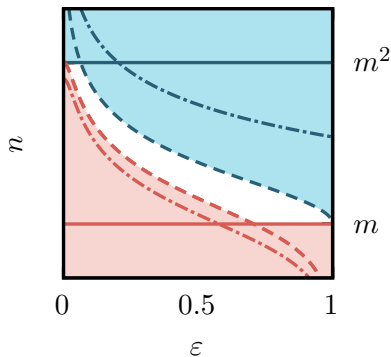
Corollary

[B 2026]

Let $m = |V| - 1$. Any set of n points and weights satisfying (MZ) on $\mathcal{F}_K$ obeys

$$n \geq \frac{m(m+1)(1-\varepsilon^2)}{2(1+m\varepsilon^2)}.$$

All bounds in one picture



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$$n = m^2$$

— dimension argument

$$n \geq m$$

--- maximal ETF construction

$$n \geq \frac{m^2(1 - \varepsilon^2)}{1 + m\varepsilon^2} \geq \frac{(m^2 - m + 1)(1 - \varepsilon^2)}{1 + (m - 1)\varepsilon^2}$$

--- graph construction

$$n \geq \frac{m(m + 1)(1 - \varepsilon^2)}{2(1 + m\varepsilon^2)}$$